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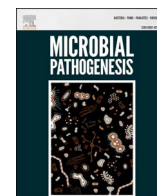


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# Characterization and resistant determinants linked to mobile elements of ESBL-producing and *mcr-1*-positive *Escherichia coli* recovered from the chicken origin

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## ABSTRACT

The spread of antimicrobial resistance (AMR) in *Escherichia coli* is a complex process linked with various mobile genetic elements (MGEs) like plasmids, transposons, and integrons. This study aimed to determine the co-occurrence of ESBL and *mcr-1* and their physical linkage with MGEs in *E. coli*. *E. coli* strains of chicken origin were obtained from different commercial farms of eastern China from 2010 to 2011. Antimicrobial sensitivity testing, identification of different antibiotic-resistant genes (ARGs), and prevalence and evidence involvement of integrons, *ISEcp1*, *ISCR1*, and *ISAp11*, were determined. A multiplex PCR was used to detect virulence genes and the phylogenetic clustering of isolates. Conjugation experiments, plasmid replicon typing were performed to know the transferability of ARGs and MGEs. A total of 83.33% of isolates were found to be multidrug-resistant (MDR). The incidence rate of *bla*<sub>CTX-M</sub>, *bla*<sub>SHV</sub>, *bla*<sub>TEM</sub>, and *mcr-1* was found to be 30%, 10.95%, 8.09%, and 36.66%, respectively. The most prevalent combination was noticed for *mcr-1* and *bla*<sub>CTX-M</sub> 73%, whereas the most prominent *bla*<sub>CTX-M</sub> alleles found, were *bla*<sub>CTX-M-55</sub> 46%, followed by *bla*<sub>CTX-M-14</sub> 31%, and *bla*<sub>CTX-M-15</sub> 13%. The frequency of *ISEcp1*, *ISCR1*, *ISAp11*, and *int1* was 27.77%, 53.70%, 51.85%, and 70.37% respectively. Most  $\beta$ -lactamases, especially *bla*<sub>CTX-M</sub>, *bla*<sub>SHV</sub>, and *bla*<sub>TEM</sub>, were associated with *ISEcp1*, *ISCR1*, and Integron 1, whereas the *ISAp11-mcr-1* segment was observed in *mcr-1*-positive *E. coli* isolates. Phylogrouping revealed that group A was the most predominant phylotype, whereas the common virulence genes detected in these isolates were EHEC, EAEC, and EPEC. Conjugation assay also indicated that multiple genetic elements were involved; common plasmids identified were FIB 61.11%, followed by IncHI2 48.14%, and FrepB 33.33%. Propagation of such MDR strains carrying multiple resistance elements among the bacterial population is a threat of worry.

## 1. Introduction

Commensal *E. coli* is a common inhabitant of the intestines of different animals and humans, and a significant pool of AMR genes that could transfer to other pathogenic microbes [1]. Some *E. coli* strains are pathogenic to both animals and humans and are documented as food-borne enteritis mediators. Pathogenic *E. coli* may be categorized into diarrheagenic *E. coli* (DEC) and extra-intestinal pathogenic *E. coli* (ExPEC) [2]. The diarrheagenic *E. coli* can be further classified into; enteroaggregative *E. coli* (EAEC), enterohemorrhagic *E. coli* (EHEC),

enteroinvasive *E. coli* (EIEC), enteropathogenic *E. coli* (EPEC), and enterotoxigenic *E. coli* (ETEC) [3,4]. Nowadays, *E. coli* infections are commonly treated with antimicrobials, which are regularly used in both animals and humans' infectious diseases [5,6]. However, the nonsystematic or misuse of antimicrobials in food animals has contributed to a significant increase of AMR in bacteria, which may be spread to humans by the intake and handling of food animals [7,8].

Bacteria can attain AMR through many mechanisms, including enzymatic inactivation of antibiotics, target modification, and dynamic efflux. Enzymatic cleavage utilizing  $\beta$ -lactamase is a powerful

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mechanism of acquired resistance against  $\beta$ -lactam antimicrobials [9]. Rise and the spread of ESBL-producing *E. coli* have been documented in humans and animals, including poultry [10], and is a serious concern that ESBLs can hydrolyze clinically significant third- and fourth-generation of cephalosporins. ESBLs can be characterized into three main groups: *bla*<sub>CTX-M</sub>, *bla*<sub>SHV</sub>, and *bla*<sub>TEM</sub> [11]. Among them, *bla*<sub>CTX-M</sub> type is the most widely reported group in ESBL-producing *Enterobacteriaceae* and is often linked with community infections [10].

Moreover, a novel mobilized colistin-resistant gene *mcr-1*, was identified for the first time in 2015 by Ref. [12] and has been documented globally in *Enterobacteriaceae*. The overwhelming diversity and transmission mechanisms of *mcr-1* between animals and humans is a major threat to global health. Our previous research has verified that the *mcr-1* gene coincides with *bla*<sub>CTX-M-15</sub> [13]. Some other studies also proved its existence with *bla*<sub>NDM</sub> [14] and other resistance genes [15], which raises concerns regarding the cure of untreatable infections worldwide.

The spread of AMR in microorganisms is a multifaceted process linked with the presence of diverse resistant determinants. Several MGEs playing a significant role in the spreading and dissemination of the ARGs among microbes [16]. Integrons, plasmids, and transposons are the critical genetic elements by which bacteria can capture and mobilize resistant genes that encode for ARGs. Different gene cassettes merged in integrons and significantly associated with AMR to  $\beta$ -lactams, aminoglycoside, and trimethoprim [17]. Of the known integrons classes, class 1 integrons have been reported in several gram-negative bacteria, including *E. coli*, *Campylobacter*, *Enterobacter*, *Klebsiella*, *Proteus*, *Salmonella*, *Shigella*, and *Vibrio* [18]. Most of these genetic elements could be more meaningful than bacterial cells in the spread of ARGs from animals to humans [19].

The incidence and role of MGEs in AMR spread in *E. coli* have been observed and confirmed worldwide. Nevertheless, little is known on the occurrence and coincidence of MGEs and their physical linkages to *mcr-1* and ESBL genes in pathogenic *E. coli* from chicken isolates. Therefore, this study aimed to determine the incidence of ESBL and *mcr-1* and their association with integrons, *ISEcp1*, *ISCR1*, and *ISAp1*, as well as plasmid typing, phylogenetic groups, and virulence factors of *E. coli* strains recovered from the chicken origin.

## 2. Materials and methods

### 2.1. Samples collection and isolation of *E. coli*

A total of 210 fecal samples showing signs of avian colibacillosis were collected in different commercial farms of eastern China from 2010 to 2011. Isolation and purification of *E. coli* were performed according to the methodology defined in our previous study [13]. The phenotypic screening was measured using double-disc diffusion assays having cefotaxime (CTX) and ceftazidime (CAZ) in the presence and absence of clavulanic acid, as per CLSI guidelines [20]. Bacterial DNA was obtained using a simple boiling method [21], and the presence of *E. coli* was confirmed by PCR. These samples were stocked in Luria-Bertani (LB) medium plus 40% glycerol at  $-80^{\circ}\text{C}$  containing 40% glycerol for further experiments.

### 2.2. Antimicrobial susceptibility testing

Antimicrobial susceptibility testing was determined using the microdilution method according to the Clinical and Laboratory Standards Institute (CLSI) [20]. A total of 13 antimicrobials representing seven antibacterial classes were used for susceptibility testing, including  $\beta$ -lactams penicillins: ampicillin (AMP), third-generation cephalosporins: cefotaxime (CTX) and ceftiofur (CEF), cephamycins: cefoxitin (CFX), carbapenems: meropenem (MEM), aminoglycosides: gentamycin (GEN), phenicols: chloramphenicol (CHL), polymyxins: colistin (COL) and polymyxin-B (PMX), quinolones: ciprofloxacin (CIP) and

enrofloxacin (ENR), sulfonamides: trimethoprim (TRI), and tetracyclines: tetracycline (TET).

### 2.3. Detection of ESBL-producing and colistin-resistant *E. coli* isolates

The ESBL-producing and colistin-resistant *E. coli* isolates were preliminarily inspected for ESBLs (*bla*<sub>CTX-M</sub>, *bla*<sub>SHV</sub>, and *bla*<sub>TEM</sub>) and colistin resistant genes *mcr-1* (1–9) by PCR technique, as described previously using specific primers [22,23]. PCR amplification was performed and sequenced to identify specific alleles of the target genes through ResFinder (<https://cge.cbs.dtu.dk/services/ResFinder/>).

### 2.4. Markers for MGEs

For the presence of MGEs, integrons and other genetic elements linked with ESBL and *mcr* genes were used as markers: class-1 integrons (*int1*), class-2 (*int2*), class-3 integrons (*int3*), *ISEcp1*, *ISCR1*, and *ISAp1*–*mcr-1* elements were screened by PCR (Table 1), as described previously using specific primers [24,25].

### 2.5. Determination of phylogenetic groups and detection of virulence genes

Phylogenetic grouping for *E. coli* is categorized into four major phylogenetic classes (A, B1, B2, and D) utilizing triplex PCR targeting five genes (*ChuA*, *yjaA*, *arpA*, *trpA*, and an unspecified DNA component (TSPE4.C2) according to Clermont et al. [26]. Isolates were further screened for the presence of various DEC virulence genes. A multiplex PCR was used to detect the virulence genes of all six categories of DEC following the conditions previously described [27].

### 2.6. Plasmid replicon typing

In order to classify plasmid incompatibility groups in ESBL- and *mcr-1* positive *E. coli* isolates, PCR-based replicon typing (PBRT) was performed using five multiplexes and three simple PCR assays, targeting twenty plasmid families (*IncHI1*, *IncHI2*, *IncI1*, *IncFIA*, *IncFIB*, *IncFIC*, *IncL/M*, *IncN*, *IncP*, *IncY*, *IncF*, *IncW*, *IncT*, *IncA/C*, *IncK*, *IncB/O*, *IncX*, *IncFIIA*, *IncI2* and *IncX4* replicons as mentioned earlier [13,28].

### 2.7. Resistance transfer assay

Conjugation experiments for 12 randomly chosen ESBL and *mcr-1* positive *E. coli* (donors) were performed to investigate the transferability of resistant genes and genetic elements from donor strains to recipient bacteria EC-600 (Nal<sup>R</sup>, Rif<sup>R</sup>). Conjugation assays were performed, as we previously defined [13]. Resistance to antimicrobial agents and further validation of the transconjugants' genetic elements, similar approaches were used as specified for respective donor strains.

## 3. Results

### 3.1. Bacterial isolation and antimicrobial susceptibility

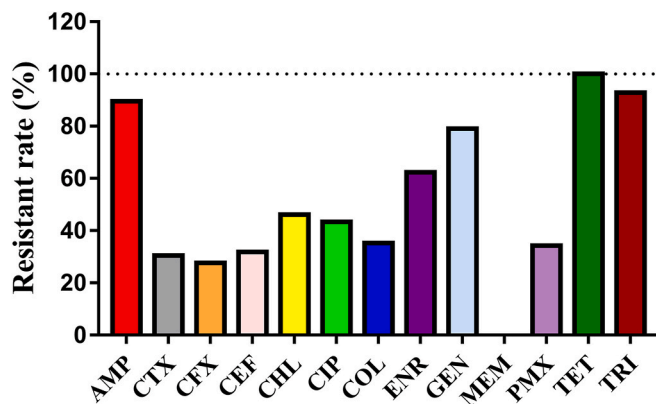
In this study, 210 *E. coli* strains were obtained from commercial broiler farms with common clinical features of avian colibacillosis from different regions of eastern China.

All strains were resistant to TET (100%), while TRI (92.85%), AMP (89.52%), GEN (79.09%), ENR (62.38%), CHL (46.19%), CIP (43.33%), CEF (31.90%), COL (35.23%), CTX (30.47%), CFX (27.61%), PMX (34.28%) and only MEM was susceptible to all of the strains (Fig. 1 and Table 2). This lower rate of meropenem is likely to be due to the fact meropenem is not used in poultry. Multidrug resistance in our isolates was found in 83.33% (Table 3). The most frequent pattern of MDR was found for tetracycline, trimethoprim, and ampicillin.

**Table 1**

Primers used for the screening of genetic elements and virulence genes.

| Target Gene            | Primer name      | 5'-3' sequence           | Annealing temp. (°C) | Product Size (bp) | References |
|------------------------|------------------|--------------------------|----------------------|-------------------|------------|
| <b>IS elements</b>     |                  |                          |                      |                   |            |
| <i>ISEcp1</i>          | <i>ISEcp1</i> -F | GTTGCTCTGTGGATAACTTG     | 55                   | 180               | [45]       |
|                        | <i>ISEcp1</i> -R | CCTAAA TTCCACGTGTGT      |                      |                   |            |
| <i>ISCR1</i>           | <i>ISCR1</i> -F  | CGCCCACTCAAACAAACG       | 55                   | 469               | [45]       |
|                        | <i>ISCR1</i> -R  | GAGGCTTTGGTGTAAACCG      |                      |                   |            |
| <i>ISApI1</i>          | <i>ISApI1</i> -F | GCGCAAATCGCAGTCG         | 58                   | 1000              | This study |
|                        | <i>ISApI1</i> -R | TGTAGGGCATTGGAGCATG      |                      |                   |            |
| <b>Integrans</b>       |                  |                          |                      |                   |            |
| <i>intI1</i>           | <i>INT-1</i> -F  | GTTCGGTCAAGGTTCTG        | 50                   | 923               | [45]       |
|                        | <i>INT-1</i> -R  | GCCAACTTTCAGCACATG       |                      |                   |            |
| <i>intI2</i>           | <i>INT-2</i> -F  | ATGTCTAACAGTCCATTTT      | 50                   | 450               | [45]       |
|                        | <i>INT-2</i> -R  | AAATCTTTAACC CGCAAAC     |                      |                   |            |
| <i>intI3</i>           | <i>INT-3</i> -F  | GCAGGGTGTGGACGAATACG     | 57                   | 760               | [45]       |
|                        | <i>INT-3</i> -R  | ACAGACCGAGAAGGCTTATG     |                      |                   |            |
| <b>Virulence genes</b> |                  |                          |                      |                   |            |
| <i>DAEC</i>            | <i>daaE</i> -F   | GAACGTTGGTTATGTGGGGTAA   | 58                   | 542               | [27]       |
|                        | <i>daaE</i> -R   | TATTCACCGGTCGGTTATCAG    |                      |                   |            |
| <i>EIEC</i>            | <i>vir</i> -F    | AGCTCAGGCAATGAAACTTTGAC  | 58                   | 618               | [27]       |
|                        | <i>vir</i> -R    | TGGGCTTGATAITCCGATAGTC   |                      |                   |            |
| <i>EAEC</i>            | <i>aafII</i> -F  | CACAGGCAACTGAAATAAGTCTGG | 58                   | 378               | [27]       |
|                        | <i>aafII</i> -R  | ATTC CATGATGTCAAGCACTTC  |                      |                   |            |
| <i>EPEC</i>            | <i>eae</i> -F    | TCAATGCAGTTCGGTTATCAGTT  | 58                   | 482               | [27]       |
|                        | <i>eae</i> -R    | GTAAAGTCCGTTACCCCAACCTG  |                      |                   |            |
| <i>EHEC</i>            | <i>stx1</i> -F   | CAGTAATGGGTGGCGAAGG      | 58                   | 348               | [27]       |
|                        | <i>stx1</i> -R   | CACAGACAATGTAACCGCTG     |                      |                   |            |
| <i>ETEC</i>            | <i>stII</i> -F   | AAAGGAGAGCTTCGTACATTTT   | 58                   | 129               | [27]       |
|                        | <i>stII</i> -R   | AATGTCGTCTTGCGTTAGGAC    |                      |                   |            |



**Fig. 1.** Antibiotic susceptibility profiles of 210 *E. coli* isolates from chicken. AMP: ampicillin, CTX: cefotaxime, CFX: cefoxitin, CEF: ceftiofur, CHL: chloramphenicol, CIP: ciprofloxacin, COL: colistin, ENR: enrofloxacin, GEN: gentamycin, MEM: meropenem, PMX: polymyxin-B, TET: tetracycline, TRI: trimethoprim.

### 3.2. Prevalence and co-existence of ESBL and *mcr-1* among *E. coli* isolates

Of the 210 *E. coli* isolates, 30% ( $n = 63/210$ ) harbored *bla*<sub>CTX-M</sub> genes, 10.95% ( $n = 23/210$ ) harbored of *bla*<sub>SHV</sub>, and 8.09% ( $n = 17/210$ ) were carrying *bla*<sub>TEM</sub> genes. Though 35.23% ( $n = 74/210$ ) of chicken *E. coli* isolates impart resistance to COL, however, the transmission rate of *mcr-1* was 36.66% ( $n = 77/210$ ). In these COL-resistant *E. coli* isolates, only the *mcr-1* gene was identified. No other COL-resistant genes *mcr* (2–9) were found in the study population of *E. coli* strains.

Based on this study's results, amongst 77 *mcr-1*-positive *E. coli* strains, 70.12% ( $n = 54/77$ ) of which were found to be carrying both *mcr-1* and ESBL genes. Distribution of *mcr-1*, in combination with other ESBL genes, showed in (Fig. 2A). The most prevalent combination was found for *mcr-1* and *bla*<sub>CTX-M</sub> (73%), followed by *mcr-1* and *bla*<sub>SHV</sub> (10%),

*mcr-1* + *bla*<sub>CTX-M</sub> + *bla*<sub>SHV</sub> (8%), *mcr-1* and *bla*<sub>TEM</sub> (5%), and *mcr-1* + *bla*<sub>CTX-M</sub> + *bla*<sub>SHV</sub> and *bla*<sub>TEM</sub> (4%). As *mcr-1* was found to be the most common in combination with *bla*<sub>CTX-M</sub> 70.12% (54/77). All these *bla*<sub>CTX-M</sub> genes were sequenced in order to identify the variants. After molecular characterization of these 54 isolated strains, it was confirmed that the majority of the *E. coli* strains harboring both *bla*<sub>CTX-M-1</sub> and *bla*<sub>CTX-M-9</sub> variants (Fig. 3). The most frequent *bla*<sub>CTX-M</sub> alleles found were *bla*<sub>CTX-M-55</sub> (46%), followed by *bla*<sub>CTX-M-14</sub> (31%), *bla*<sub>CTX-M-15</sub> (13%), *bla*<sub>CTX-M-28</sub> (6%), and *bla*<sub>CTX-M-66</sub> (4%) (Fig. 2B).

### 3.3. Molecular traits of resistant *E. coli* strains co-harboring *mcr-1* and ESBL genes

Based on this study's results, most of the *mcr-1* positive *E. coli* strains carrying *bla*<sub>CTX-M</sub> genes. Further genetic features showed that 64.81% ( $n = 35/54$ ) of the *mcr-1*-positive *E. coli* strains referred to the *bla*<sub>CTX-M-1</sub> group, followed by the *bla*<sub>CTX-M-9</sub> group 31.48% ( $n = 17/54$ ) (Fig. 3). Twenty strains 37.03% ( $n = 20/54$ ) had *mcr-1* and *bla*<sub>TEM</sub> genes, while 29.62% ( $n = 16/54$ ) of *E. coli* strains having *mcr-1* and *bla*<sub>SHV</sub> genes. The presence of MGEs, which play a critical role in the spreading ESBLs and *mcr-1* genes, were investigated. On evaluation analysis of MGEs displayed that 27.77% ( $n = 15/54$ ), 53.70% ( $n = 29/54$ ), 51.85% ( $n = 28/54$ ), and 70.37% ( $n = 38/54$ ) of the *E. coli* strains carried *ISEcp1*, *ISCR1*, *ISApI1*, and integrans, respectively (Fig. 4). Majority of the  $\beta$ -lactamases, mainly those encoding *bla*<sub>CTX-M</sub>, *bla*<sub>SHV</sub>, and *bla*<sub>TEM</sub>, were found physically linked with *ISEcp1*, *ISCR1*, and Integrin 1. The most prevalent combination observed in *mcr-1*-positive *E. coli* strains was the *ISApI1*-*mcr-1* segment 51.80% ( $n = 28/54$ ) (Table 4). The phylogenetic screening of the selected strains confirmed that phylo-group A was the commonly observed phenotype 77.77% ( $n = 42/54$ ), followed by other less common groups including, group B1 7.40% ( $n = 4/54$ ), group B2 9.25% ( $n = 5/54$ ), and phylo-group D 9.25% ( $n = 5/54$ ). The findings of the virulence genes profiles are presented in (Fig. 3). The most prevalent DEC *E. coli* strains found were EHEC 59.25% ( $n = 32/54$ ), followed by EAEC 12.96% ( $n = 7/54$ ), and EPEC 9.25% ( $n = 5/54$ ) (Fig. 4). All ESBL and *mcr-1*-positive *E. coli* strains presented plasmids of wide range replicon types. The most common replicon types identified were IncFIB

**Table 2**Distribution of MICs of thirteen antimicrobials against MDR *E. coli* isolates ( $n = 210$ ).

| Antibiotics     | MIC (mg/L) |      |     |    |    |    |    |    |    |    |     |     |      | Resistance |
|-----------------|------------|------|-----|----|----|----|----|----|----|----|-----|-----|------|------------|
|                 | <0.125     | 0.25 | 0.5 | 1  | 2  | 4  | 8  | 16 | 32 | 64 | 128 | 256 | >256 |            |
| Ampicillin      | -          | -    | -   | 0  | 2  | 4  | 6  | 10 | 6  | 22 | 39  | 62  | 59   | 89.52%     |
| Cefotaxime      | 16         | 13   | 38  | 46 | 33 | 11 | 14 | 5  | 3  | 8  | 13  | 5   | 5    | 30.47%     |
| Cefoxitin       | -          | 13   | 47  | 36 | 26 | 13 | 17 | 8  | 19 | 8  | 17  | 4   | 2    | 27.61%     |
| Ceftiofur       | 2          | 8    | 35  | 47 | 51 | 15 | 8  | 17 | 5  | 9  | 8   | 3   | 2    | 31.90%     |
| Chloramphenicol | -          | -    | 1   | 4  | 5  | 41 | 23 | 39 | 25 | 19 | 22  | 17  | 14   | 46.19%     |
| Ciprofloxacin   | 6          | 16   | 28  | 27 | 42 | 13 | 20 | 31 | 14 | 8  | 2   | 3   | -    | 43.33%     |
| Colistin        | 31         | 29   | 41  | 35 | 37 | 21 | 9  | 3  | 2  | 2  | 0   | -   | -    | 35.23%     |
| Enrofloxacin    | 5          | 11   | 24  | 8  | 16 | 15 | 29 | 43 | 31 | 14 | 12  | 2   | -    | 62.38%     |
| Gentamycin      | 4          | 6    | 13  | 7  | 4  | 8  | 2  | 27 | 53 | 32 | 21  | 17  | 16   | 79.09%     |
| Meropenem       | 151        | 22   | 11  | 9  | 17 | 0  | 0  | 0  | 0  | -  | -   | -   | -    | 00.00 %    |
| Polymyxin-B     | 37         | 15   | 22  | 64 | 18 | 29 | 11 | 9  | 3  | 2  | 0   | -   | -    | 34.28%     |
| Tetracycline    | -          | -    | 0   | 0  | 0  | 0  | 0  | 21 | 15 | 26 | 51  | 73  | 24   | 100.0%     |
| Trimethoprim    | -          | -    | 0   | 0  | 2  | 1  | 0  | 7  | 5  | 14 | 77  | 62  | 42   | 92.85%     |

Notes: Red vertical lines are the cutoff points between intermediate and resistant values. White areas display a number of dilutions tested for each antimicrobial agent.

**Table 3**Multidrug-resistant patterns of chicken *E. coli* isolates.

| No. of antibiotics | Resistance pattern                                    | No. of isolates | Resistance (%) |
|--------------------|-------------------------------------------------------|-----------------|----------------|
| 3                  | TET, TRI, AMP                                         | 175             | 83.33          |
| 4                  | TET, TRI, AMP, GEN                                    | 145             | 69.04          |
| 5                  | TET, TRI, AMP, GEN, CHL                               | 119             | 56.66          |
| 6                  | TET, TRI, AMP, GEN, CHL, CIP                          | 75              | 35.71          |
| 7                  | TET, TRI, AMP, CEF, CTX, CIP, CEF                     | 39              | 18.57          |
| 8                  | TET, TRI, AMP, GEN, ENR, CHL                          | 16              | 7.61           |
| 9                  | CIP, CEF                                              | 3               | 1.42           |
| 10                 | TET, TRI, AMP, GEN, CEF, CTX                          | 1               | 0.47           |
| 11                 | COL, CFX, PMX                                         | 1               | 0.47           |
|                    | TET, TRI, AMP, GEN, ENR, CHL, CIP, CTX, COL, PMX      |                 |                |
|                    | TET, TRI, AMP, GEN, CFX, ENR, CHL, CEF, COL, PMX, CTX |                 |                |

**Abbreviations:** AMP: ampicillin, CTX: cefotaxime, CFX: cefoxitin, CEF: ceftiofur, CHL: chloramphenicol, CIP: ciprofloxacin, COL: colistin, ENR: enrofloxacin, GEN: gentamycin, MEM: meropenem, PMX: polymyxin-B, TET: tetracycline, TRI: trimethoprim.

61.11% ( $n = 33/54$ ), followed by IncHI2 48.14% ( $n = 26/54$ ), IncFrepB 33.33% ( $n = 18/54$ ), IncFIC 27.77% ( $n = 27/54$ ), IncX4 27.77% ( $n = 27/54$ ), IncFIA 14.81% ( $n = 8/54$ ), IncP 11.11% ( $n = 6/54$ ), IncN 7.40% ( $n = 4/54$ ), IncY 7.40% ( $n = 4/54$ ), and IncA/C 5.55% ( $n = 3/54$ ) (Fig. 3).

### 3.4. ESBLs and *mcr-1* may be conjugatively transferred by plasmids and MGEs

Conjugative plasmids trigger the transmission of different ARGs and genetic elements. Randomly 12 *mcr-1* and ESBL-positive *E. coli* strains were used in conjugation assays. Several resistant genes and genetic factors linked were passed from donor strains to recipient *E. coli* EC600 in conjugation assays (Table 5). Out of 12 strains, nine strains were found by carrying different genes and genetic elements. Conjugation assay verified that several genetic factors were carried by narrow-range plasmids such as IncFIB, IncHI2. The conjugative plasmid types found in our experiments which confer resistant genes from donor to recipient *E. coli* were, IncF, IncHI2, and IncX4 type replicons. However, various genes combination found to be physically linked could not be shifted with all resistance traits from the donor to the recipient bacteria.

## 4. Discussion

Antimicrobial use in food-producing animals is a challenge for public health as resistant bacteria can be developed and spread to humans by ingesting and handling these foodstuffs [29]. The current study was performed on chicken isolates to investigate the prevalence rate of ESBL-producing and COL-resistant *E. coli* strains and also to examined the presence of MGEs and their role in resistance and virulence of chicken origin's.

Overall, the incidence of ESBL-producing and *mcr-1*-positive *E. coli*

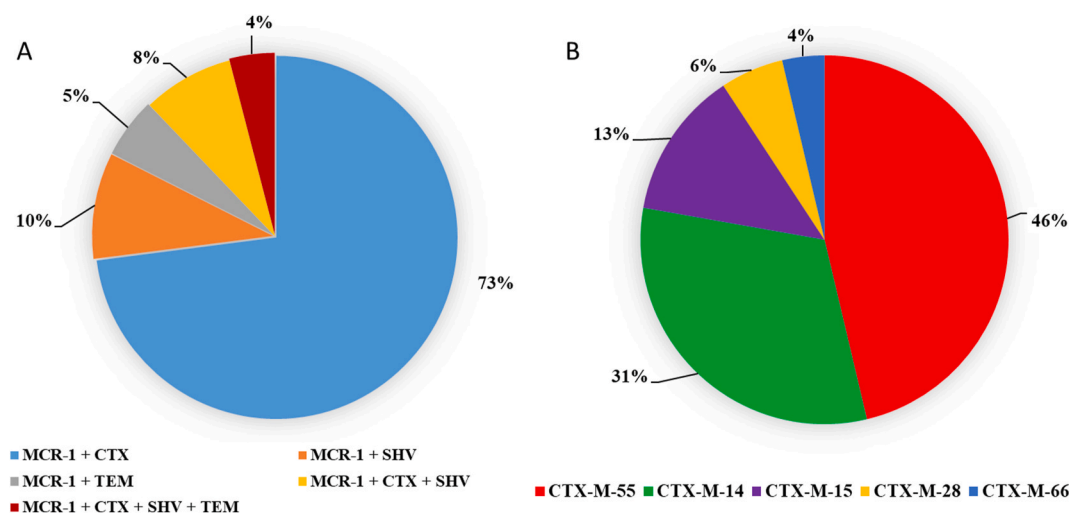


Fig. 2. Distribution of various resistance genes in combination (A) and CTX types (B).

was less in chicken, as we previously reported in pigs and cattle studies [22,23]; however, the prevalence of AMR among isolates was high. Our results suggested that both positive ESBL-producing and *mcr-1*-positive isolates presented MDR (83.33%). *E. coli* strains were classified as MDR according to the classifications established by Magiorakos et al. [30]. MDR observed in this study was less than that of our previous findings in cattle (85%) and swine (100%). It may be due to the limited number of chicken isolates, which were isolated from 2010–11. Similarly, the resistance rate of colistin was also higher in swine (70.18%) and cattle (71.07%) than that in chicken (35.23%) isolates of our study. This major disparity may be clarified by the fact that far more colistin has been used in the pig and dairy industries as a feed additive to support animal growth in China. Previous research from China showed that colistin resistance and the *mcr-1* gene were linked with higher colistin use in *E. coli* isolates from poultry as early as in the 1980s. However, this gene's identification rate has rapidly increased from 5.2% in 2009 to 30% in 2014, which may explain the findings of this research [31]. Furthermore, MDR phenotypes' existence revealed that the co-occurrence of ESBLs with other resistant traits in *E. coli* isolates results in the development of their susceptibility range to  $\beta$ -lactams and other antimicrobial agents, which is a significant problem for the usage of antimicrobials in poultry production. Therefore, ESBL *E. coli* can be used as an indicator for screening MDR *E. coli* in food-producing animals.

Among 77 *mcr-1*-positive *E. coli* strains, (70.12%) were identified to harbor both *mcr-1* and ESBL genes. The previous studies reported that *bla*<sub>CTX-M</sub> is the most predominant ESBL-encoding gene found in the ESBL-positive *E. coli* isolates [32–36]. The prevalent genotype of *bla*<sub>CTX-M</sub> detected in this study was *bla*<sub>CTX-M-55</sub>, are in line with the previous studies [34,37–39]. The frequency of *bla*<sub>CTX-M-55</sub> was greater than *bla*<sub>CTX-M-14</sub> and *bla*<sub>CTX-M-15</sub>, while the next most prevalent *bla*<sub>CTX-M</sub> found was *bla*<sub>CTX-M-14</sub>, which surpassed *bla*<sub>CTX-M-15</sub>. Several recent studies have confirmed that *bla*<sub>CTX-M-55</sub> is the second most prevalent *bla*<sub>CTX-M</sub> in human isolates in China, which was not reported in human clinical isolates prior to 2010 [34,40]. Interestingly, the frequency of *bla*<sub>CTX-M-55</sub> was higher in combination with *mcr-1* (46%), which is somewhat close to the latest study on chicken origin in China [37].

The development of AMR in bacteria is a dynamic process involving a number of pathways [41]. MGEs such as plasmids and transposons with resistant genes playing a significant role in propagating and circulating these resistant determinants between the various strains [42, 43]. The current study is unique in that it analyses MGEs and the frequencies of their prevalence, specifically, with ESBL and *mcr-1*-positive *E. coli* strains. The most typical genetic elements observed were class 1 integrons (70.37%), followed by *ISCR1* (53.70%), *ISAp11* (51.85%), and *ISEcp1* (27.77%). Most of the  $\beta$ -lactamases, particularly those encoding

*bla*<sub>CTX-M-55</sub>, *bla*<sub>CTX-M-14</sub>, and *bla*<sub>CTX-M-15</sub>, were found physically linked to *ISCR1*, *ISEcp1*, and class 1 Integron. Similar reports for *bla*<sub>CTX-M-14</sub> and *bla*<sub>CTX-M-15</sub> have been observed in previous researches [44,45]. Unlike previous studies, here we reported that majority of *bla*<sub>CTX-M-55</sub> genes were physically linked to the *ISCR1*, *ISEcp1*, and class 1 Integron, which was missing in the earlier studies. In this report, *ISAp11* transposon co-harboring with *mcr-1* was found in a number of *E. coli* strains (51.85%) probably involved in horizontal transfer and apparently a key element resulting in the extensive spreading of the *mcr-1* gene; nevertheless, the *ISAp11-mcr-1* fragment may be lost due to intense recombination [46]. In all *mcr-1* harboring strains, the *ISAp11-mcr-1* fragment could not have been identified. Other studies have also shown that the *mcr-1* gene could also be mobilized with other MGEs or exist independently [47–49]. Propagation of such multiple genetic components, carrying a range of different resistant determinants, could be responsible for the co-resistance against multi drugs observed amongst our strains.

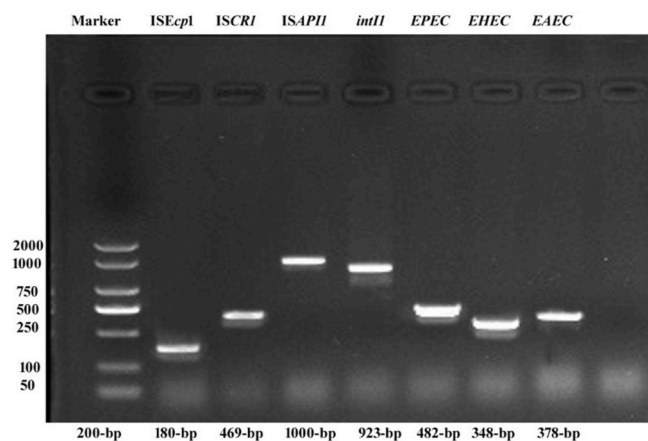
The phylogenetic findings showed that ESBL and *mcr-1*-harboring *E. coli* strains belonged to the phylogenetic group A (77.7%), with some strains of group B1, B2, and D observed. Parallel findings of various DEC strains, the most prevalent virulence factor found was EHEC (59.2%), which were similar to the previous report in Eastern China [50]. Conversely, the phylogroup found in our research was group A while group D in another study. Furthermore, conjugation experiments have confirmed that multiple resistance genes and genetic elements have been passed from donors to recipients with narrow and wide-ranging plasmids such as IncFIB, IncHI2, and IncX4. Conjugative plasmids (especially plasmids containing IncF-type, IncHI2, and IncX4 replicons) of our chicken isolates were parallel to those in strains from other studies [45,51,52]. Overall, this study suggests that the pervasive ESBL-producing and COL-resistant *E. coli* harboring *mcr-1* in chicken isolates includes numerous plasmid and genetic factor clones.

## 5. Conclusion

The most predominant combination was found for *mcr-1* and *bla*<sub>CTX-M</sub> (73%). The most prevalent *bla*<sub>CTX-M</sub> alleles detected were *bla*<sub>CTX-M-55</sub> (46%), followed by *bla*<sub>CTX-M-14</sub> in (31%), and *bla*<sub>CTX-M-15</sub> in (13%). Prevalence of *ISEcp1*, *ISCR1*, *ISAp11*, and *int1* was 27.77%, 53.70%, 51.85%, and 70.37% respectively. This study reports carriage of MDR *E. coli* strains with multiple genetic elements and virulent factors involved. Such wide dissemination of these resistance genes with MGEs in chicken-origin commensal bacteria is a major public health concern for preventing and treating MDR *E. coli*.

| Strain | ESBLs          |   |   |   |    |   |    | blaTEM | blaSHV | mcr-1 | ISEcp1 | ISCR1 | ISAp1-mcr-1 | Integron 1 | Phylo-group |   |      |      | Virulence type(s) |        |        | Plasmid replicon type(s) |        |        |      |      |      |          |       |  |  |
|--------|----------------|---|---|---|----|---|----|--------|--------|-------|--------|-------|-------------|------------|-------------|---|------|------|-------------------|--------|--------|--------------------------|--------|--------|------|------|------|----------|-------|--|--|
|        | blaCTX-M group |   |   |   |    | A | B1 |        |        |       |        |       |             |            | B2          | D | EHEC | EAEC | EPEC              | IncA/C | IncFIA | IncFIB                   | IncFIC | IncHI2 | IncP | IncN | IncY | IncFrepB | IncX4 |  |  |
|        | 1              | 2 | 8 | 9 | 25 |   |    |        |        |       |        |       |             |            |             |   |      |      |                   |        |        |                          |        |        |      |      |      |          |       |  |  |
| EC-07  |                |   |   |   |    |   |    |        |        |       |        |       |             |            |             |   |      |      |                   |        |        |                          |        |        |      |      |      |          |       |  |  |
| EC-08  |                |   |   |   |    |   |    |        |        |       |        |       |             |            |             |   |      |      |                   |        |        |                          |        |        |      |      |      |          |       |  |  |
| EC-11  |                |   |   |   |    |   |    |        |        |       |        |       |             |            |             |   |      |      |                   |        |        |                          |        |        |      |      |      |          |       |  |  |
| EC-13  |                |   |   |   |    |   |    |        |        |       |        |       |             |            |             |   |      |      |                   |        |        |                          |        |        |      |      |      |          |       |  |  |
| EC-17  |                |   |   |   |    |   |    |        |        |       |        |       |             |            |             |   |      |      |                   |        |        |                          |        |        |      |      |      |          |       |  |  |
| EC-19  |                |   |   |   |    |   |    |        |        |       |        |       |             |            |             |   |      |      |                   |        |        |                          |        |        |      |      |      |          |       |  |  |
| EC-21  |                |   |   |   |    |   |    |        |        |       |        |       |             |            |             |   |      |      |                   |        |        |                          |        |        |      |      |      |          |       |  |  |
| EC-22  |                |   |   |   |    |   |    |        |        |       |        |       |             |            |             |   |      |      |                   |        |        |                          |        |        |      |      |      |          |       |  |  |
| EC-23  |                |   |   |   |    |   |    |        |        |       |        |       |             |            |             |   |      |      |                   |        |        |                          |        |        |      |      |      |          |       |  |  |
| EC-34  |                |   |   |   |    |   |    |        |        |       |        |       |             |            |             |   |      |      |                   |        |        |                          |        |        |      |      |      |          |       |  |  |
| EC-38  |                |   |   |   |    |   |    |        |        |       |        |       |             |            |             |   |      |      |                   |        |        |                          |        |        |      |      |      |          |       |  |  |
| EC-41  |                |   |   |   |    |   |    |        |        |       |        |       |             |            |             |   |      |      |                   |        |        |                          |        |        |      |      |      |          |       |  |  |
| EC-42  |                |   |   |   |    |   |    |        |        |       |        |       |             |            |             |   |      |      |                   |        |        |                          |        |        |      |      |      |          |       |  |  |
| EC-44  |                |   |   |   |    |   |    |        |        |       |        |       |             |            |             |   |      |      |                   |        |        |                          |        |        |      |      |      |          |       |  |  |
| EC-49  |                |   |   |   |    |   |    |        |        |       |        |       |             |            |             |   |      |      |                   |        |        |                          |        |        |      |      |      |          |       |  |  |
| EC-51  |                |   |   |   |    |   |    |        |        |       |        |       |             |            |             |   |      |      |                   |        |        |                          |        |        |      |      |      |          |       |  |  |
| EC-52  |                |   |   |   |    |   |    |        |        |       |        |       |             |            |             |   |      |      |                   |        |        |                          |        |        |      |      |      |          |       |  |  |
| EC-55  |                |   |   |   |    |   |    |        |        |       |        |       |             |            |             |   |      |      |                   |        |        |                          |        |        |      |      |      |          |       |  |  |
| EC-58  |                |   |   |   |    |   |    |        |        |       |        |       |             |            |             |   |      |      |                   |        |        |                          |        |        |      |      |      |          |       |  |  |
| EC-59  |                |   |   |   |    |   |    |        |        |       |        |       |             |            |             |   |      |      |                   |        |        |                          |        |        |      |      |      |          |       |  |  |
| EC-60  |                |   |   |   |    |   |    |        |        |       |        |       |             |            |             |   |      |      |                   |        |        |                          |        |        |      |      |      |          |       |  |  |
| EC-62  |                |   |   |   |    |   |    |        |        |       |        |       |             |            |             |   |      |      |                   |        |        |                          |        |        |      |      |      |          |       |  |  |
| EC-67  |                |   |   |   |    |   |    |        |        |       |        |       |             |            |             |   |      |      |                   |        |        |                          |        |        |      |      |      |          |       |  |  |
| EC-69  |                |   |   |   |    |   |    |        |        |       |        |       |             |            |             |   |      |      |                   |        |        |                          |        |        |      |      |      |          |       |  |  |
| EC-70  |                |   |   |   |    |   |    |        |        |       |        |       |             |            |             |   |      |      |                   |        |        |                          |        |        |      |      |      |          |       |  |  |
| EC-73  |                |   |   |   |    |   |    |        |        |       |        |       |             |            |             |   |      |      |                   |        |        |                          |        |        |      |      |      |          |       |  |  |
| EC-74  |                |   |   |   |    |   |    |        |        |       |        |       |             |            |             |   |      |      |                   |        |        |                          |        |        |      |      |      |          |       |  |  |
| EC-79  |                |   |   |   |    |   |    |        |        |       |        |       |             |            |             |   |      |      |                   |        |        |                          |        |        |      |      |      |          |       |  |  |
| EC-81  |                |   |   |   |    |   |    |        |        |       |        |       |             |            |             |   |      |      |                   |        |        |                          |        |        |      |      |      |          |       |  |  |
| EC-82  |                |   |   |   |    |   |    |        |        |       |        |       |             |            |             |   |      |      |                   |        |        |                          |        |        |      |      |      |          |       |  |  |
| EC-84  |                |   |   |   |    |   |    |        |        |       |        |       |             |            |             |   |      |      |                   |        |        |                          |        |        |      |      |      |          |       |  |  |
| EC-85  |                |   |   |   |    |   |    |        |        |       |        |       |             |            |             |   |      |      |                   |        |        |                          |        |        |      |      |      |          |       |  |  |
| EC-88  |                |   |   |   |    |   |    |        |        |       |        |       |             |            |             |   |      |      |                   |        |        |                          |        |        |      |      |      |          |       |  |  |
| EC-89  |                |   |   |   |    |   |    |        |        |       |        |       |             |            |             |   |      |      |                   |        |        |                          |        |        |      |      |      |          |       |  |  |
| EC-90  |                |   |   |   |    |   |    |        |        |       |        |       |             |            |             |   |      |      |                   |        |        |                          |        |        |      |      |      |          |       |  |  |
| EC-93  |                |   |   |   |    |   |    |        |        |       |        |       |             |            |             |   |      |      |                   |        |        |                          |        |        |      |      |      |          |       |  |  |
| EC-94  |                |   |   |   |    |   |    |        |        |       |        |       |             |            |             |   |      |      |                   |        |        |                          |        |        |      |      |      |          |       |  |  |
| EC-98  |                |   |   |   |    |   |    |        |        |       |        |       |             |            |             |   |      |      |                   |        |        |                          |        |        |      |      |      |          |       |  |  |
| EC-99  |                |   |   |   |    |   |    |        |        |       |        |       |             |            |             |   |      |      |                   |        |        |                          |        |        |      |      |      |          |       |  |  |
| EC-113 |                |   |   |   |    |   |    |        |        |       |        |       |             |            |             |   |      |      |                   |        |        |                          |        |        |      |      |      |          |       |  |  |
| EC-115 |                |   |   |   |    |   |    |        |        |       |        |       |             |            |             |   |      |      |                   |        |        |                          |        |        |      |      |      |          |       |  |  |
| EC-118 |                |   |   |   |    |   |    |        |        |       |        |       |             |            |             |   |      |      |                   |        |        |                          |        |        |      |      |      |          |       |  |  |
| EC-138 |                |   |   |   |    |   |    |        |        |       |        |       |             |            |             |   |      |      |                   |        |        |                          |        |        |      |      |      |          |       |  |  |
| EC-147 |                |   |   |   |    |   |    |        |        |       |        |       |             |            |             |   |      |      |                   |        |        |                          |        |        |      |      |      |          |       |  |  |
| EC-150 |                |   |   |   |    |   |    |        |        |       |        |       |             |            |             |   |      |      |                   |        |        |                          |        |        |      |      |      |          |       |  |  |

**Fig. 3.** Genetic features of 54 ESBL- and COL-resistant *E. coli* isolates. The figure presents grey squares which indicate the identified gene, insertion sequences, phylogroups, virulence genes, and replicon types.



**Fig. 4.** Screening of mobile genetic elements and virulence genes in *E. coli* using PCR assay.

**Table 4**

Analysis of physical linkages of ESBL and *mcr-1* with various genetic elements.

| Resistance genes               | No. of isolates screened | Number (%) of ESBLs and <i>mcr-1</i> physically linked to various mobile genetic elements |              |               |              |
|--------------------------------|--------------------------|-------------------------------------------------------------------------------------------|--------------|---------------|--------------|
|                                |                          | <i>ISEcp1</i>                                                                             | <i>ISCR1</i> | <i>ISApI1</i> | Integrations |
| <i>bla</i> <sub>CTX-M-55</sub> | 25                       | 14                                                                                        | 9            | 18            | 21 (84.0)    |
| <i>bla</i> <sub>CTX-M-14</sub> | 17                       | (56.0)                                                                                    | (36.0)       | (72.0)        | 15 (88.2)    |
| <i>bla</i> <sub>CTX-M-15</sub> | 7                        | 8 (47.0)                                                                                  | 12           | 13            | 2 (28.5)     |
| <i>bla</i> <sub>CTX-M-28</sub> | 3                        | 3 (42.8)                                                                                  | (70.5)       | (76.4)        | 2 (66.6)     |
| <i>bla</i> <sub>CTX-M-66</sub> | 2                        | 1                                                                                         | 5            | 5 (71.4)      | 2 (100)      |
| <i>bla</i> <sub>SHV</sub>      | 16                       | (33.33)                                                                                   | (71.4)       | 1             | 8 (50.0)     |
| <i>bla</i> <sub>TEM</sub>      | 20                       | 0                                                                                         | 0            | (33.33)       | 13 (65.0)    |
| <i>mcr-1</i>                   | 54                       | 6 (37.5)                                                                                  | 1            | 1 (50.0)      | 38 (70.3)    |
|                                |                          | 4 (20.0)                                                                                  | (50.0)       | 9 (56.2)      |              |
|                                |                          | 15                                                                                        | 11           | 14            |              |
|                                |                          | (27.7)                                                                                    | (68.7)       | (70.0)        |              |
|                                |                          |                                                                                           | 10           | 28            |              |
|                                |                          |                                                                                           | (50.0)       | (51.8)        |              |
|                                |                          |                                                                                           | 29           |               |              |
|                                |                          |                                                                                           | (53.7)       |               |              |

**Table 5**  
Horizontal transfer of resistance genes and related genetic features.

| Strains  | CTX<br>MIC<br>(mg/L) | COL<br>MIC<br>(mg/L) | MGEs or ARGs detected in<br>donors and recipients                                                                                         | Plasmid replicons<br>detected   |
|----------|----------------------|----------------------|-------------------------------------------------------------------------------------------------------------------------------------------|---------------------------------|
| D-EC-07  | 128                  | 8                    | <i>mcr-1</i> ; <i>bla</i> <sub>CTX-M-15</sub> ; <i>ISEcp1</i> ;<br><i>ISCR1</i> ; <i>ISAP11</i> ; <i>int1</i>                             | IncFIB, IncHI2                  |
| T-EC-07  | 128                  | 4                    | <i>mcr-1</i> ; <i>bla</i> <sub>CTX-M-15</sub> ; <i>ISCR1</i> ;<br><i>ISAP11</i> ; <i>int1</i>                                             | IncHI2                          |
| D-EC-11  | 64                   | 4                    | <i>mcr-1</i> ; <i>bla</i> <sub>CTX-M-55</sub> ; <i>bla</i> <sub>SHV</sub> ;<br><i>ISEcp1</i> ; <i>ISCR1</i> ; <i>ISAP11</i> ; <i>int1</i> | IncHI2, IncFrepB                |
| T-EC-11  | 128                  | 8                    | <i>mcr-1</i> ; <i>bla</i> <sub>CTX-M-55</sub> ; <i>ISEcp1</i>                                                                             | IncHI2                          |
| D-EC-34  | 256                  | 2                    | <i>mcr-1</i> ; <i>bla</i> <sub>CTX-M-14</sub> ; <i>bla</i> <sub>TEM</sub> ;<br><i>ISCR1</i> ; <i>ISAP11</i> ; <i>int1</i>                 | IncFIA, IncFIB,<br>IncN, IncX4  |
| T-EC-34  | 32                   | 4                    | <i>mcr-1</i> ; <i>bla</i> <sub>CTX-M-14</sub> ; <i>ISCR1</i> ;<br><i>ISAP11</i>                                                           | IncFIA, IncFIB,<br>IncN, IncX4  |
| D-EC-42  | >256                 | 16                   | <i>ISCR1</i> ; <i>ISAP11</i> ; <i>int1</i> ; <i>mcr-1</i> ;<br><i>bla</i> <sub>CTX-M-55</sub> ; <i>bla</i> <sub>SHV</sub>                 | IncFIB, IncFIC,<br>IncHI2       |
| T-EC-42  | 16                   | 0.5                  | <i>bla</i> <sub>CTX-M-55</sub> ; <i>ISCR1</i> ; <i>int1</i>                                                                               | IncFIB, IncFIC,<br>IncHI2       |
| D-EC-49  | 128                  | 16                   | <i>mcr-1</i> ; <i>bla</i> <sub>CTX-M-15</sub> ; <i>ISEcp1</i> ;<br><i>ISCR1</i> ; <i>ISAP11</i> ; <i>int1</i>                             | IncHI2, IncY                    |
| T-EC-49  | 256                  | 1                    | <i>bla</i> <sub>CTX-M-15</sub> ; <i>ISCR1</i> ; <i>int1</i>                                                                               | IncHI2, IncY                    |
| D-EC-55  | 256                  | 2                    | <i>mcr-1</i> ; <i>bla</i> <sub>CTX-M-55</sub> ; <i>ISEcp1</i> ;<br><i>ISAP11</i> ; <i>int1</i>                                            | IncFIB, IncHI2                  |
| T-EC-55  | 64                   | 4                    | <i>mcr-1</i> ; <i>bla</i> <sub>CTX-M-55</sub> ;<br><i>ISEcp1</i> ; <i>ISAP11</i>                                                          | IncFIB, IncHI2                  |
| D-EC-85  | 128                  | 2                    | <i>mcr-1</i> ; <i>bla</i> <sub>CTX-M-14</sub> ; <i>ISEcp1</i> ;<br><i>ISAP11</i> ; <i>int1</i>                                            | IncFIB, IncFIC,<br>IncHI2, IncP |
| T-EC-85  | 1                    | 8                    | <i>mcr-1</i> ; <i>ISAP11</i>                                                                                                              | IncFIB, IncFIC,<br>IncHI2, IncP |
| D-EC-94  | 128                  | 4                    | <i>mcr-1</i> ; <i>bla</i> <sub>CTX-M-14</sub> ; <i>ISEcp1</i> ;<br><i>ISAP11</i> ; <i>int1</i>                                            | IncFIB                          |
| T-EC-94  | 64                   | 8                    | <i>mcr-1</i> ; <i>bla</i> <sub>CTX-M-14</sub> ;<br><i>ISEcp1</i> ; <i>ISAP11</i>                                                          | IncFIB                          |
| D-EC-147 | 32                   | 2                    | <i>mcr-1</i> ; <i>bla</i> <sub>CTX-M-55</sub> ; <i>bla</i> <sub>SHV</sub> ;<br><i>ISEcp1</i> ; <i>ISAP11</i> ; <i>int1</i>                | IncFIB, IncFIC,<br>IncX4        |
| T-EC-147 | 64                   | 4                    | <i>mcr-1</i> ; <i>bla</i> <sub>CTX-M-55</sub> ; <i>bla</i> <sub>SHV</sub> ;<br><i>ISEcp1</i> ; <i>int1</i>                                | IncFIB, IncFIC,<br>IncX4        |
| D-EC-150 | 128                  | 16                   | <i>mcr-1</i> ; <i>bla</i> <sub>CTX-M-14</sub> ; <i>ISEcp1</i> ;<br><i>ISCR1</i> ; <i>ISAP11</i> ; <i>int1</i>                             | IncFIB, IncFIC                  |
| T-EC-150 | 256                  | 32                   | <i>mcr-1</i> ; <i>bla</i> <sub>CTX-M-14</sub> ; <i>ISCR1</i> ;<br><i>ISAP11</i> ; <i>int1</i>                                             | IncFIB, IncFIC                  |
| D-EC-164 | 32                   | 8                    | <i>mcr-1</i> ; <i>bla</i> <sub>CTX-M-66</sub> ; <i>bla</i> <sub>TEM</sub> ;<br><i>ISEcp1</i> ; <i>ISCR1</i> ; <i>ISAP11</i>               | IncFIB, IncHI2,<br>IncFrepB     |
| T-EC-164 | 4                    | 0.25                 | <i>bla</i> <sub>CTX-M-66</sub> ; <i>bla</i> <sub>TEM</sub> ; <i>ISCR1</i>                                                                 | IncFIB, IncHI2,<br>IncFrepB     |
| D-EC-179 | 256                  | 2                    | <i>mcr-1</i> ; <i>bla</i> <sub>CTX-M-55</sub> ; <i>bla</i> <sub>SHV</sub> ;<br><i>ISCR1</i> ; <i>ISAP11</i> ; <i>int1</i>                 | IncX4                           |
| T-EC-179 | 256                  | 4                    | <i>mcr-1</i> ; <i>bla</i> <sub>CTX-M-55</sub> ; <i>bla</i> <sub>SHV</sub> ;<br><i>ISCR1</i> ; <i>ISAP11</i>                               | IncX4                           |
| EC-600   | 0.25                 | 0.125                | Not determined                                                                                                                            | Not determined                  |

The transmitted MGEs and ARGs to the transconjugants are shown in bold.

**CRedit authorship contribution statement**

**Muhammad Shafiq:** Conceptualization, Methodology, Data curation, Writing - original draft. **Liping Wang:** Supervision, Writing - review & editing.

**Declaration of competing interest**

The authors report no conflicts of interest in this work.

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